

JavaScript Array Methods Assignment

Part A: Multiple Choice Questions

1. What will be the output of the following code?

```
let arr = [10, 20, 30];  
console.log(arr.length);
```

- a) 2
- b) 3
- c) 4
- d) undefined

2. Which method adds an element to the end of an array? a) unshift()

- b) shift()
- c) push()
- d) pop()

3. What does the `pop()` method return? a) The modified array

- b) The removed element
- c) The array length
- d) undefined

4. Which loop prints the index of array elements? a) `for...of`

- b) `for...in`
- c) `while`
- d) `do...while`

5. What will be the output?

```
let arr = [1, 2, 3];
```

```
let arr2 = [...arr];  
console.log(arr2);
```

- a) [1, 2, 3]
- b) [[1, 2, 3]]
- c) undefined
- d) Error

Part B: Short Answer Questions

1. Write the output of the following code:

```
let numbers = [5, 10, 15, 20];  
console.log(numbers[1]);  
console.log(numbers[4]);
```

2. What is the difference between (shift()) and (unshift()) methods?

3. Write code to create an array using the Array constructor with values 100, 200, 300.

4. Explain what happens when you access a negative index in a JavaScript array.

5. Write the output:

```
let fruits = ['apple', 'banana', 'orange'];  
fruits.reverse();  
console.log(fruits);
```

Part C: Code Writing

1. Create an array of 5 student names and add a new student at the beginning using the appropriate method.

2. Write code to remove the last two elements from this array: `['HTML', 'CSS',`

'JavaScript', 'React',

'Node'

3. Create two arrays and combine them using:

Spread operator

- concat() method

4. Write a for loop to print all elements of an array in reverse order without using the (reverse()) method.

5. Create an array of numbers and write code to:

- Add 100 at the end
- Add 0 at the beginning
- Remove the first element
- Print the final array

Part D: Problem Solving

1. Write a program that:

- Creates an array of 5 browser names
- Uses a for...of loop to find if 'Chrome' exists
- If found, print "Chrome browser found!" and stop the loop

2. Create a function that takes an array of words and returns a sentence by joining them with spaces.

Test with: (['I', 'love', 'JavaScript'])

3. Write a program that:

- Creates an array of employee data with mixed data types (name, age, salary, isActive)
- Prints each element with its index using for...in loop
- Format: "Index 0: John"

4. Create a shopping cart system:

- Start with an empty array
- Add 3 items using push()
- Remove the last item
- Add an item at the beginning
- Join all items with commas and print as "Shopping List: item1, item2, item3"

Answer Sheet

Part A: Multiple Choice Answers

1. b) 3

Arrays have a length property that returns the number of elements

2. c) push()

push() adds elements to the end of an array

3. b) The removed element

pop() removes and returns the last element

4. b) for...in

for...in loop iterates over indices/keys

5. a) [1, 2, 3]

Spread operator creates a shallow copy of the array

Part B: Short Answer Solutions

1. Output:

10

undefined

`numbers[1]`) returns the second element
(10)

`numbers[4]`) is undefined as the array has only 4 elements (indices 0-3)

2. Difference between `shift()` and `unshift()`:

`shift()`: Removes and returns the first element from the array

`unshift()`: Adds one or more elements to the beginning of the array

3. Code:

```
let numbers = new Array(100, 200, 300);
```

4. Negative index in JavaScript arrays:

JavaScript doesn't support negative indexing

Accessing a negative index returns (`undefined`)

- Unlike Python, negative indices don't count from the end

5. Output:

```
['orange', 'banana', 'apple']
```

Part C: Code Writing Solutions

1. Adding student at beginning:

```
let students = ['Alice', 'Bob', 'Charlie', 'David', 'Eve'];
students.unshift('Frank');
console.log(students); // ['Frank', 'Alice', 'Bob', 'Charlie', 'David', 'Eve']
```

2. Removing last two elements:

```
let techs = ['HTML', 'CSS', 'JavaScript', 'React', 'Node'];
techs.pop(); // removes 'Node'
techs.pop(); // removes 'React'
console.log(techs); // ['HTML', 'CSS', 'JavaScript']
```

3. Combining arrays:

```
// Using spread operator
let arr1 = [1, 2, 3];
let arr2 = [4, 5, 6];
let combined1 = [...arr1, ...arr2];
console.log(combined1); // [1, 2, 3, 4, 5, 6]
```

```
// Using concat()
let combined2 = arr1.concat(arr2);
console.log(combined2); // [1, 2, 3, 4, 5, 6]
```

4. Reverse printing without reverse():

```
let colors = ['red', 'green', 'blue', 'yellow'];
for (let i = colors.length - 1; i >= 0; i--) {
  console.log(colors[i]);
}
```

// Output: yellow, blue, green, red

5. Array operations:

```
let numbers = [10, 20, 30, 40, 50];  
numbers.push(100);  
                // Add 100 at the end  
  
numbers.unshift(0);  
numbers.shift();  
                // Add 0 at the beginning  
                // Remove first element  
  
console.log(numbers); // [10, 20, 30, 40, 50, 100]
```

Part D: Problem Solving Solutions

1. Browser search program:

```
let browsers = ['Firefox', 'Safari', 'Chrome', 'Opera', 'Edge'];  
for (let browser of browsers) {  
  if (browser === 'Chrome') {  
    console.log('Chrome browser found!');  
    break;  
  }  
}
```

2. Word joining function:

```
function createSentence (wordsArray) {  
  
}  
  return wordsArray.join(' ');
```

```
let words = ['I', 'love', 'JavaScript'];  
let sentence = createSentence (words);  
console.log(sentence); // "I love JavaScript"
```

3. Employee data with for...in:

```
let empData = ['John', 25, 50000, true];
for (let index in empData) {

}
console.log('Index ${index}: ${empData[index]}');
```

// Output:

//Index 0: John

// Index 1: 25

//Index 2: 50000

// Index 3: true

4. Shopping cart system:

```
let shoppingCart = [];
// Add 3 items
shoppingCart.push('Milk');
shoppingCart.push('Bread');
shoppingCart.push('Eggs');
// Remove last item
shoppingCart.pop();
// Add item at beginning
shoppingCart.unshift('Coffee');
// Join and print
let shoppingList = shoppingCart.join(', ');
console.log('Shopping List: ${shoppingList}');
// Output: "Shopping List: Coffee, Milk, Bread"
```

Additional Practice Tips

1. Always remember that JavaScript arrays are zero-indexed
2. Array methods like `push()` and `unshift()` can accept multiple arguments
3. Methods like `pop()` and `shift()` can only remove one element at a time
4. The spread operator (`...`) is useful for copying and combining arrays
5. Different loops serve different purposes:

- Regular for loop: When you need the index

- `for...of`: When you need the values

- `for...in`: When you need the keys/indices